TABLEAU · LESSON 1 OF 6

Getting Started with Tableau: Interface, Data Connections and Your First Views

Understand the Tableau product family, install Tableau Public, connect to the Superstore India data, read the Data Source page, learn dimensions vs measures and discrete vs continuous fields, master the Marks card and shelves, build and sort your first views, and save them to Tableau Public.

✦ Tableau Desktop / Tableau Public 2026 | Beginner | ⏱ 120 min

By the end of this lesson you can

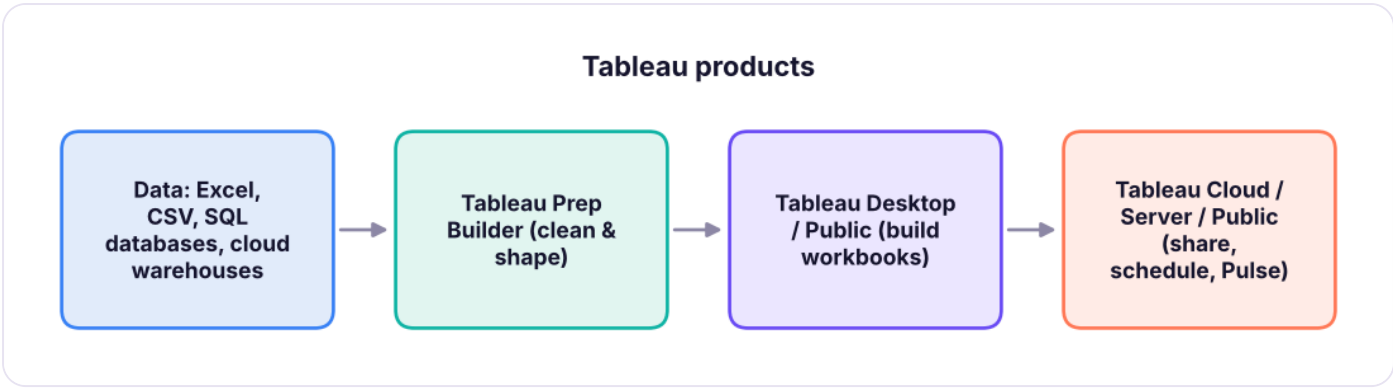
- Describe Tableau Desktop, Public, Cloud, Server, Prep and Pulse, and when each is used
- Connect to a CSV, check data types and rename or hide fields on the Data Source page
- Explain dimensions vs measures and discrete (blue) vs continuous (green) fields
- Build bar, line and text-table views with the Columns, Rows, Marks, Filters and Pages shelves
- Sort, label, format and save a workbook to Tableau Public

Datasets used

`superstore_sales.csv`

What is Tableau?

Tableau is a visual analytics platform: you drag fields onto shelves and Tableau writes the query, aggregates the data and draws the chart. It is owned by Salesforce and, with Power BI, is one of the two BI tools most often named in Indian analyst job posts. Where Power BI leans on DAX formulas and the Microsoft stack, Tableau is known for fast visual exploration and highly customisable charts.



The Tableau product family. You build in Desktop (or Public), share on Cloud/Server/Public, and prepare data in Prep.

Product	What it is	Cost (2026)
Tableau Public (desktop app + website)	Free version of Desktop; workbooks are saved **publicly** on public.tableau.com	Free
Tableau Desktop	Full authoring tool; saves locally, connects to databases	Paid (Creator licence)
Tableau Cloud	Tableau's hosted service for sharing, refresh schedules and permissions	Paid, per user
Tableau Server	The same as Cloud, but installed on your company's own servers	Paid
Tableau Prep Builder	Visual tool to clean, join and reshape data before analysis	Included with Creator
Tableau Pulse / Tableau Agent	AI features: metric digests in plain language and an assistant that helps write calculations and build views	Cloud add-ons

TIP **Which one should you install?**

Learners should install **Tableau Public** (free). It has almost every authoring feature of Desktop. The two limits: workbooks are saved to your public profile (so never use confidential data) and it connects mainly to files (Excel, CSV, text, JSON, PDF, spatial) and Google Sheets rather than company databases. Your public profile also works as a portfolio for interviews.

Install and connect

STEP BY STEP **Install Tableau Public and connect to Superstore**

1. Go to public.tableau.com → **Create** → download **Tableau Public** for Windows or Mac and install it; sign up for a free profile.
2. Open Tableau Public. On the **Connect** pane choose **Text file** and pick `superstore_sales.csv` (for an Excel file choose **Microsoft Excel**).
3. The **Data Source page** opens: the table appears in the canvas and a preview grid below it.
4. Check each column's **data type icon** (Abc = string, # = number, calendar = date, globe = geographic). Click an icon to change it.
5. Click **Sheet 1** at the bottom to start building.

The data you connect to is the same Superstore India file used in the Excel and Power BI lessons. Check its size before you build anything:

Python

```python
import pandas as pd
df = pd.read_csv("superstore_sales.csv", parse_dates=["order_date", "ship_date"])
print(df.shape)
print(df.dtypes.astype(str).value_counts().to_dict())
print("Orders from", df.order_date.min().date(), "to", df.order_date.max().date())
df.head(3)
```

OUTPUT

```
(6323, 19)
{'object': 12, 'float64': 3, 'int64': 2, 'datetime64[ns]': 2}
Orders from 2023-01-01 to 2025-12-31
```

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name
0	1	IN-2023-00001	2023-01-01	2023-01-05	Second Class	C0765	Ishaan Gupta
1	2	IN-2023-00001	2023-01-01	2023-01-05	Second Class	C0765	Ishaan Gupta
2	3	IN-2023-00001	2023-01-01	2023-01-05	Second Class	C0765	Ishaan Gupta

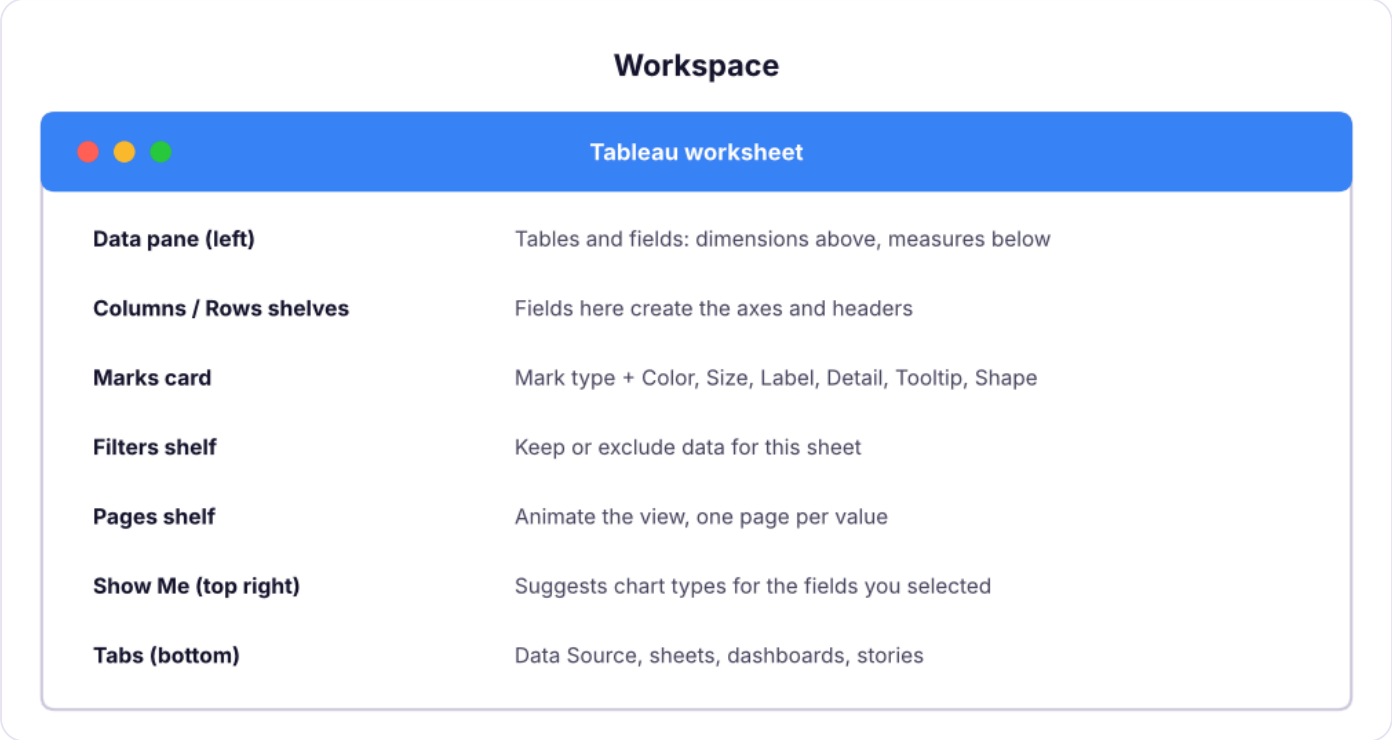
3 rows × 19 columns

6,323 rows (order lines) and 19 columns: Tableau will show these as **fields** in the Data pane.

The Data Source page

Feature	Where	Use
Rename a field	Click the column header → Rename	Friendly names such as "Sub-Category"
Hide a field	Column menu → Hide	Remove IDs you will never use
Change type / geographic role	Click the type icon	Make `state` a **State/Province** so it maps
Split	Column menu → Split	`customer_name` → first and last name
Pivot	Select columns → Pivot	Turn month columns into rows
Data Interpreter	Left pane checkbox	Clean messy Excel sheets with titles and merged cells
Live vs Extract	Top right	Extract (.hyper) is a fast snapshot; Live queries the source every time
Filters (data source)	Top right → Add	Keep only the rows the workbook needs

The workspace

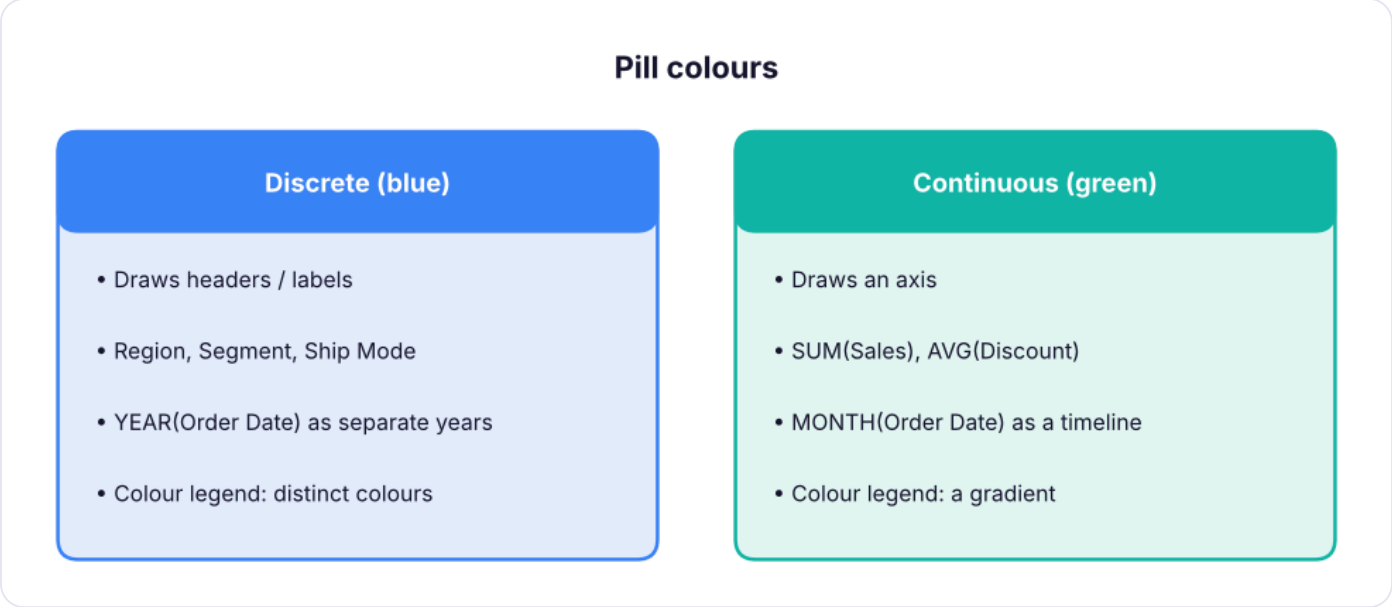


The Tableau worksheet layout. Drag fields from the Data pane onto shelves and the Marks card.

Dimensions vs measures, discrete vs continuous

Tableau sorts every field in two ways, and the colour of a pill tells you which:

	Meaning	Example	What it creates
Dimension	Categories you slice by	Region, Category, Customer Name	Groups (one mark per value)
Measure	Numbers you aggregate	Sales, Profit, Quantity	SUM, AVG, COUNT...
Discrete (blue pill)	Separate values	Region, YEAR(Order Date) as "2023, 2024"	**Headers**
Continuous (green pill)	An unbroken range	SUM(Sales), exact date	**Axes**



Blue (discrete) pills create headers; green (continuous) pills create axes.

A measure dropped into the view is **aggregated** (default SUM). Right-click the pill → **Measure** to switch to Average, Median, Count, Count (Distinct), Minimum, Maximum, Standard deviation and so on.

Your first views

Sales by region (bar chart)

Drag **Region** to Rows and **Sales** to Columns. Tableau writes the equivalent of `SELECT region, SUM(sales) FROM superstore GROUP BY region` and draws one bar per region:

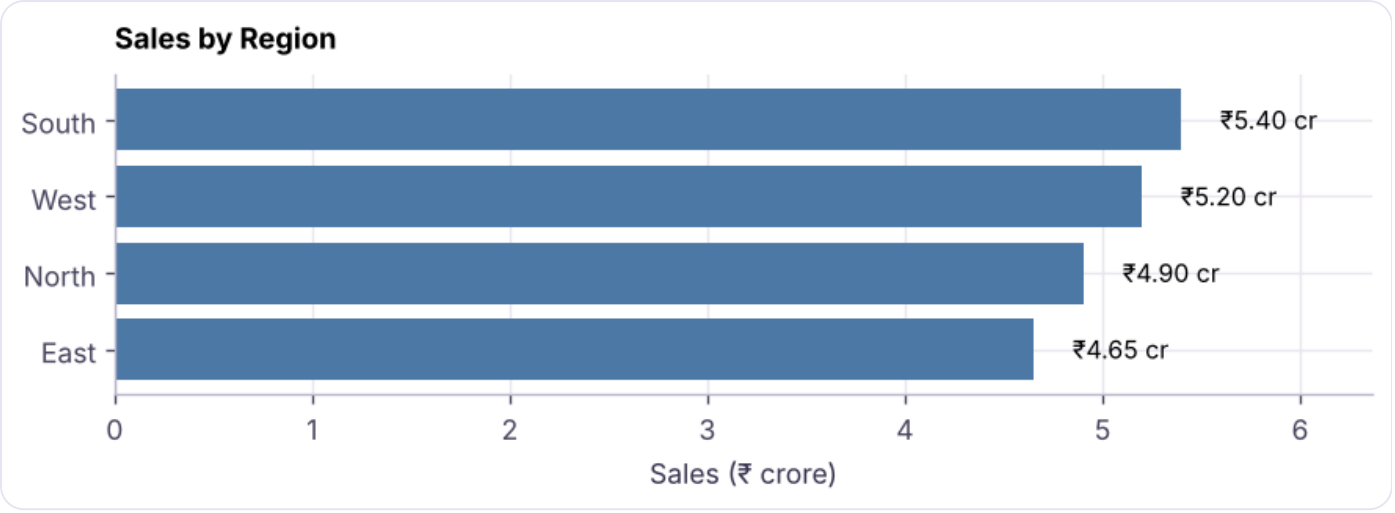
Python

```
by_region = (df.groupby("region")[["sales", "profit"]].sum() /
1e7).round(2).sort_values("sales", ascending=False)
by_region.columns = ["Sales (₹ crore)", "Profit (₹ crore)"]
by_region
```

OUTPUT

	Sales (₹ crore)	Profit (₹ crore)
region		
South	5.40	0.47
West	5.20	0.47
North	4.90	0.46
East	4.65	0.44

4 rows × 2 columns



The view you get: SUM(Sales) by Region, sorted descending, with labels.

The default Tableau colours (blue #4E79A7, orange #F28E2B, red #E15759…) are the **Tableau 10** palette used in the figures in this track.

STEP BY STEP **Sort, label and colour it**

1. Hover the Sales axis and click the **sort** icon, or click the toolbar sort descending button.
2. Click **Label** on the Marks card → tick *Show mark labels*.
3. Drag **Profit** onto **Color**: bars are shaded by profit, so sales and margin show together.
4. Right-click the Sales axis → **Format** → Numbers → Currency (Custom) → Display units: *Crores* is not built in, so use a calculated field `[Sales]/1e7` named "Sales (₹ cr)" (Lesson 3), or show *Millions*.
5. Double-click the sheet tab and rename it "Sales by Region".

Sales over time (line chart)

Drag **Order Date** to Columns: Tableau shows **YEAR(Order Date)** as a discrete header. Click the **+** on the pill to drill down to Quarter and Month. Right-click the pill and choose the **green** "Month" (continuous) to draw a single timeline, or the **blue** "Month" (discrete) to compare the same month across years.

Python

```
m = df.set_index("order_date").sales.resample("ME").sum() / 1e5
print("Months:", len(m), "| best month:", m.idxmax().strftime("%b %Y"), f"₹{m.max():.1f}
lakh", "| weakest:", m.idxmin().strftime("%b %Y"), f"₹{m.min():.1f} lakh")
yearly = (df.groupby(df.order_date.dt.year).sales.sum() / 1e7).round(2); yearly
```

OUTPUT

```
Months: 36 | best month: Nov 2025 ₹102.8 lakh | weakest: Jan 2023 ₹24.5 lakh
```

	sales
order_date	
2023	5.76
2024	6.54
2025	7.84

3 rows × 1 columns

Text table (crosstab)

Put **Category** on Rows, **YEAR(Order Date)** on Columns and **Sales** on **Text**. Add **Analysis → Totals → Show Row Grand Totals**. This is Tableau's pivot table:

Python

```
ct = pd.crosstab(df.category, df.order_date.dt.year, values=df.sales, aggfunc="sum",
margins=True, margins_name="Grand Total") / 1e5
ct.round(1)
```

OUTPUT

order_date	2023	2024	2025	Grand Total
category				
Furniture	99.5	121.3	127.6	348.4
Office Supplies	26.1	35.5	39.7	101.3
Technology	450.8	497.2	616.5	1,564.4
Grand Total	576.4	654.0	783.8	2,014.1

4 rows × 4 columns

Show Me

Show Me (top right) highlights the chart types that fit the fields you have selected and states what it needs, e.g. "*1 date, 1 or more dimensions, 1 or more measures*" for a line chart. It is a fast start, but learn to build each chart by hand: interviewers often ask *how* you would build a view.

Saving and sharing

Format	What it contains
`.twb` workbook	Sheets, dashboards and the connection, **no data**
`.twbx` packaged workbook	Workbook **plus** a copy of the data/extract: share this one
`.hyper` extract	Tableau's compressed, fast data snapshot
Tableau Public profile	**File → Save to Tableau Public**; anyone can view and download it

COMMON MISTAKES **Beginner mistakes**

- Saving confidential company data to Tableau Public: everything there is public.
- Sending a `.twb` file: the receiver cannot open it without your data. Send `.twbx`.
- Leaving a date as a string ("Abc"): you lose drill-down and date functions. Change the type on the Data Source page.
- Using a measure that should be a dimension (e.g. Postal Code as SUM): right-click → Convert to Dimension.

2026 UPDATE

Tableau in 2026

- **Tableau Agent** (formerly Einstein Copilot for Tableau) suggests calculations and builds views from a typed question in Tableau Cloud.
- **Tableau Pulse** sends plain-language metric summaries ("Sales are up 8% this week, driven by the West region") to email and Slack.
- **Tableau Next** is Salesforce's agent-based analytics layer on top of Data Cloud. Desktop and Public remain the tools you learn first.

Summary

KEY TAKEAWAYS

- Tableau turns drag-and-drop into queries: shelves define the structure, the Marks card defines the look.
- Dimensions slice, measures aggregate; blue pills are discrete (headers), green pills are continuous (axes).
- Start with Tableau Public, connect to a file, fix data types on the Data Source page, then build.
- Share packaged workbooks (`.twbx`) or publish to Tableau Public/Cloud.

Quick quiz

6 questions · tap an answer to check it

1. What does a green pill on Columns create?

A A continuous axis

B A set of headers

C A filter

D A calculated field

Answer: A. Green = continuous = axis; blue = discrete = headers.

2. You drag Sales into the view. What does Tableau show by default?

- **A** SUM(Sales) for each combination of the dimensions in the view
- **B** Every row's sales value
- **C** The average sales
- **D** The count of rows

Answer: A. Measures are aggregated, and SUM is the default.

3. Which file should you email a colleague so they can open your work with the data?

- **A** A packaged workbook (.twbx)
- **B** A workbook (.twb)
- **C** A .hyper file only
- **D** A screenshot

Answer: A. .twbx bundles the workbook with its data.

4. What is the main limitation of Tableau Public?

- **A** Workbooks are saved publicly and it connects mainly to files
- **B** It cannot draw maps
- **C** It has no calculated fields
- **D** It only works on Mac

Answer: A. Never put confidential data on Tableau Public.

5. State is shown as "Abc" and will not draw a map. What do you do?

- **A** Assign the geographic role State/Province
- **B** Convert it to a measure
- **C** Delete the field
- **D** Use a dual axis

Answer: A. Geographic roles give a field latitude and longitude.

6. Which feature suggests chart types for the selected fields?

- **A** Show Me
- **B** Pages shelf
- **C** Data Interpreter
- **D** Extract

Answer: A. Show Me lists the chart types and what each needs.

EXERCISE **Practice**

Connect to `superstore_sales.csv` in Tableau Public and build three sheets: (1) Sales by Category sorted descending with labels; (2) a continuous monthly Sales line with Profit on Color; (3) a text table of Segment × Year with grand totals. Save as a packaged workbook and publish to your Tableau Public profile.

Show solution (worked answer)

Sheet 1: Category on Rows, SUM(Sales) on Columns, sort descending, Label on. Sheet 2: MONTH(Order Date) continuous (green) on Columns, SUM(Sales) on Rows, SUM(Profit) on Color. Sheet 3: Segment on Rows, YEAR(Order Date) on Columns, SUM(Sales) on Text, Analysis → Totals → Show Row and Column Grand Totals. File → Save to Tableau Public.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between a dimension and a measure? A dimension is a categorical field used to group the data (Region, Category); a measure is a numeric field that Tableau aggregates (SUM of Sales). Each mark represents one combination of the dimensions in the view.

Q: Discrete vs continuous? Discrete fields (blue) create headers and distinct colours; continuous fields (green) create axes and colour gradients. Dates can be either: discrete YEAR gives "2023, 2024" headers, continuous MONTH gives a timeline.

Q: .twb vs .twbx vs .hyper? .twb is the workbook definition without data; .twbx packages the workbook with its data or extract; .hyper is the extract file itself.

Q: Live connection or extract? Live always shows current data but depends on the source's speed; an extract is a fast, compressed snapshot that you refresh on a schedule. Use extracts for speed and offline work, live for real-time needs or very large governed warehouses.

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